UNIT-4

NETWORK LAYER

Network Layer:-

The Network Layer is the third layer of the OSI model that is responsible for delivering data (packets) from the source to the destination across multiple networks.

Need of Network Layer

The Network Layer is required because:-

1. End to-End Delivery - Transfers data from source to destination across multiple networks.
2. Logical Addressing - Uses IP Address to uniquely identify devices.
3. Routing :- Finds the best path among many possible paths.

4. Internetworking :- Connects different types of network (LAN, WAN)

5. Handling Large Networks :- Enables communication over long distances.

Services Provided by Network Layer

1. Connectionless service (Datagram)
   - No prior connection setup
   - Each packet is treated independently
   - Example :- Internet (IP)

2. Connection - Oriented service (Virtual Circuit)
   - Path is established before transmission
   - All packets follow the same route.

Routing Algorithm

A routing algorithm is a method used by routers to determines the best path for sending data (packets) from source to destination in a network.

Types of Routing Algorithm

1. Static Routing
   - Routes are manually configured
   - Does not change automatically
   - Simple but not suitable for large networks
2. Dynamic Routing
   - Routes are updated automatically
   - Adapts to network changes
   - Used in real-world networks

Main Routing Algorithm.

1. Least ~~Count~~ Cost Routing Algorithm.
2. Dijkstra's Algorithm.
3. Bellman-ford algorithm.
4. Hierarchical Routing.
5. Broadcast Routing.
6. Multicast Routing.

Least - Cost Routing Algorithm (LCR)

LCR is a routing technique used in CN to finds the minimum cost path from a source node to a destination node.

"Cost" can means Distance and Time delay

Goal → Send data through the most efficient (lowest cost) path

Common algorithm used → Dijkstra's Algorithm (used in Link state Routing).

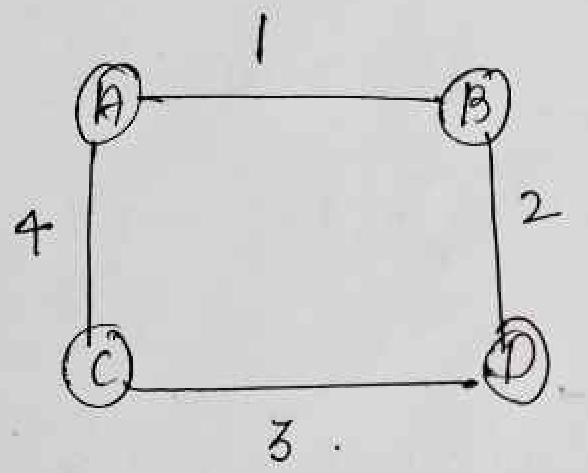
Dijkstra's Algorithm

1. Initialize.
   - set distance of source node = 0
   - set distance of all other nodes ∞.
2. Mark all nodes as unvisited
3. Select the nodes with the smallest tentative distance
4. update distance of its neighbors
   - New distance = current distance + edge cost
   - if smaller → update it.

5. Mark node as visited
   - Once visited, it won't be checked again
6. Repeat until all nodes are visited.

Simple Example.

Suppose we have node.



find shortest path A to D

- $A \to B \to D \to 1 + 2 = 3$ (least cost)
- $A \to C \to D = 4 + 3 = 7$

So, best path = $A \to B \to D$

Key features

- Find shortest path efficiently
- Used in Link state Routing Protocols.
- Work with weighted graphs
- Guarantees optimal solution.

Bellman-Ford Algorithm

2024

The Bellman-Ford algorithm is a shortest path algorithms used to find the shortest distance from a single source vertex to all other vertices in a weighted graph.

Unlike Dijkstra's Algorithm, Bellman Ford can handle negative edge weights and can also detect negative weight cycle.

Algorithm Steps

Given a Graph with V vertices and E edges.

1 Initialize distances:

Distance ~~of~~ source = 0

Distance of all other vertices = ∞

2. Relax all edges V-1 times

- for every edge (u, v) with weight w.

  if dist [u] + w < dist [v]

  dist [v] = dist [u] + w.

3. Check for negative cycles.

Relax all edges one more time
If any distance can still be
reduced, a negative weight cycle
exists.

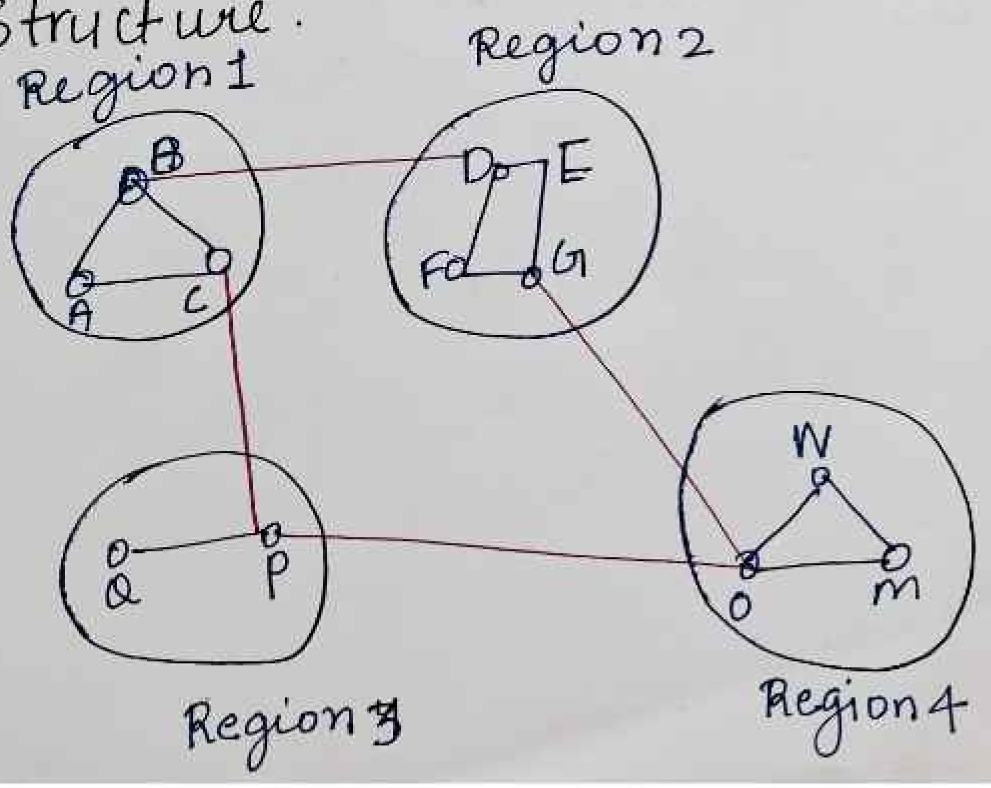
Hierarchical Routing -

- Hierarchical Routing is a routing technique in which a large network is divided into smaller regions (areas, zones. or domains).
- Routers maintain detailed information about their own region and summarized. information about other regions.

Why it is used?

- Reduces routing table size
- Decreases routing update traffic
- Improves scalability in large network
- Makes network management easier

Structure.



Each Router knows:-

- Complete routes within its own area.
- Only summarized routes for other areas

Advantages

1. Smaller routing tables
2. Faster routing decisions
3. Suitable for large network

Disadvantages.

1. More complex design
2. May not always provide the shortest path.
3. Requires careful planning.

Broadcast Routing.

Broadcast routing is a networking technique used to deliver a packet from one source to all nodes in a network.

Common approaches include:-

① Flooding → Each router forwards the packet on every outgoing link except the one it arrived on.

- Simple but can create duplicate packets and waste bandwidth

② Controlled Flooding – uses mechanisms such as sequence numbers or hop limits (TTL) to reduce duplicates and loop.

③ Spanning Tree Broadcast Routers form a loop-free tree structure.

Broadcast packets are forwarded only along the tree branches; ensuring each node receives the packet exactly once.

4. Reverse Path Forwarding (RPF)

- A router forwards a broadcast packet only ~~along the tree bro~~ if it arrives on the interface that would normally be used to reach the source.

- Helps prevent loops and unnecessary transmissions.

Uses.

- Routing protocol updates
- Service discovery (e.g. finding devices on a local network)
- Network management and announcements

Drawbacks.

- High bandwidth consumption.
- Potential broadcast storms.
- Limited scalability in large networks

Multicast Routing -

- Multicast routing is a method of sending data from one sender to a selected group of receivers (multicast group) instead of sending it to every device.

- Multicast routing ~~devices~~ delivers packets onto to hosts that have joined a specific multicast group, making communication more bandwidth - efficient than broadcast.

How its works

1. A sender transmits data to a multicast group address.
2. Receivers join the multicast group
3. Routers creates a multicast distribution tree.
4. Packets are forwarded only to routers and hosts that belongs to the group.

Advantages :-

1 Save network bandwidth
2 Reduce sender workload
3 Efficient for one-to-many communication.

Applications

- Live video streaming
- Online lectures / webinars
- Video confrencing.
- Real- time stock updates

IP Address

An IP Address (Internet Protocol address) is a unique numerical identifier assigned to the every device connected to a network.

It helps to:

- Identify the device
- Locate a device on a network.

Just like a home address helps deliver letters. an IP address helps deliver data.

Types of IP Addresses.

① IPv4 (Internet Protocol version 4)

② IPv6 (Internet Protocol version 6)

IPv4 Imp

IPv4 (Internet Protocol Version 4) is the fourth version of the Internet Protocol used to identify devices on a network and enable communication between them.

Format of IPv4

- An IPV4 address consists of 4 numbers (octets) separated by dots (.).
- Each number ranges from 0 to 255.

Example →
192.168.1.1
10.0.0.5
172.16.0.1

Structure of IPV4

IPV4 uses 32 bits divided into 4 parts

192 . 168 . 1 . 1
11000000 . 10101000 . 00000001 . 00000001

Each part = 8 bits (1 byte)

Total = 8+8+8+8 = 32 bits.

Features of IPV4.

1. Uses 32-bit addressing
2. Supports about 4.3 billion unique addresses.
3. Easy to understand and implement
4. Widely used on the Internet.

Advantages.

- Simple and widely supported
- Easy network configuration
- Compatible with most devices.

Disadvantages.

- Limited number of addresses.
- Address shortage due to increasing internet
- users.
- Less secure than IPv6.

IPV6 Imp

IPV6 (Internet Protocol Version 6) is the latest version of the Internet Protocol.
It was developed to overcome the shortage of IPv4 addresses and provide a much larger number of unique addresses for devices connected to the Internet.

format at IPv6

- IPv6 addresses are written in 8 groups of hexadecimal numbers.
- Groups are separated by colons (:).
- Hexadecimal digits include 0-9 and A-F.

Example>

2001 : 0db8 : 85a3 : 0000 : 0000 :
8a2e : 0370 : 7334

Structure of IPv6.

- IPv6 uses 128 bits.
- Divided into 8 blocks, each block contains 16 bits.

2001 : 0db8 : 85a3 : 0000 : 0000 : 8a2e :
0370 : 7334

Total = 128 bits.

Features of IPv6

1. Uses 128-bit addressing
2. Provides a huge number of unique IP addresses.
3. Better security support through IPsec.
4. Faster and more efficient routing
5. Supports automatic address configuration.

Advantages

- Almost unlimited IP addresses.
- Improved security
- Better network performance.
- Supports mordern Internet applications and IoT devices.

Disadvantages.

- More complex than IPv4.
- Requires new hardware / software support in some cases
- Migration from IPv4 can be costly

IP Header format. Imp

The IP Header format contains Information required to route a packet from source to destination.

Version	Length(IHL) (IHL)	Type of service (TOS)	Total length	
Identification			Flag	Fragment Offset
Time to Live (TTL)	Protocol		Header Checksum	
Source IP Address				
Destination IP Address				
Options.				

Field Explanation.

1. Version (4 bits) → IP version (IPv4 = 4)
2. IHL (4 bits) → Internet Header Length.
3. Type of Service (8 bits) - Priority and QoS information.
4. Total length (16 bits) - Total size of packet.

5. Identification (16 bits)- Identifies fragments of a packet.

6. Flag (3 bits)- Controls fragmentation.

7. Fragment Offset (13 bits)→ Position of fragment in original packet.

8 TTL (8 bits)→ Time to Live :- prevents packets from looping forever

9. Protocol (8-bits)- Indicates upper-layer protocol (TCP, UDP, ICMP).

10. Header Checksum (16 bits)- Error checking of header.

11. Source Address (32 bits)→ Sender's IP address.

12. Destination Address (32 bits)→ Receiver's IP Address.

13. Options → Optional control Information.

14- Data → Actual message being transmitted.

PACKET FORWARDING

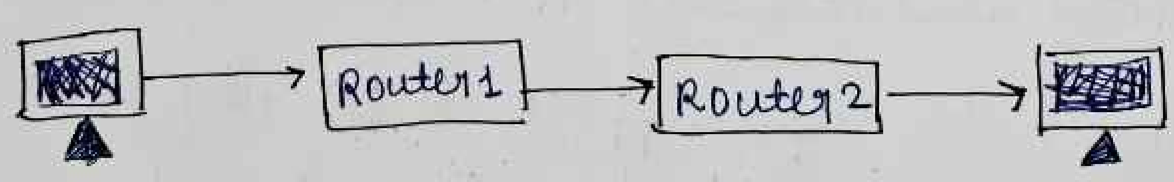
Packet Forwading is the process ~~p~~ of sending a packet from a router's input interface to the correct output interface so that it reaches its destination.

How Packet Forwarding Works.

1. A packet arrives at a router
2. The router reads the Destination IP Address from the packet.
3. The router checks its forwarding Table (Routing Table).
4. It finds the best route for the destination.
5. The packet is forwarded through the appropriate output port.
6. This process continue until the packet reaches the destination.

Example

Suppose a packet is sent from Computer A to Computer B.



Router 1 :- checks the destination address and forwards the packet to Router 2

Router 2 :- again checks the destination and forwards it to computer B.

Advantages

- Faster delivery of packets
- Efficient use of network resources.
- Helps packets reach the correct destination.

Fragmentation.

2024

Fragmentation is the process of dividing a large IP packet into smaller pieces (fragments) so that it can travel through a network whose Maximum Transmission Unit (MTU) is smaller than the packet size.

Why fragmentation is Needed.

Different networks support different packet sizes. If a packet is too large for a network, it must be broken into smaller fragments.

Example → Suppose a packet size is 4000 bytes. but the network allows only 1500 bytes per packet.

Original Packet (4000 bytes).

Fragmentation

↓

Fragment 1 = 1500 bytes.

Fragment 2 = 1500 bytes.

Fragment 3 = 1000 bytes.

These fragments are sent separately through the network.

Advantages

- Allows transmission through networks with smaller MTU.
- Improves compatibility between different networks.

ICMP 2024, 2025, 2023

ICMP (Internet Control Message Protocol) is a network layer protocol used to send error messages and network status information between devices on an IP network.

It helps computers and routers identify and report network problems.

Why ICMP is Needed.

When a packet cannot reach its destination ICMP informs the sender about the problem.

- Destination is unreachable.
- Packet's time limit has expired.
- Router is congested.
- Network connection problem

Function of ICMP.

1) Error Reporting → IMCP reports error that occurs during packet transmission.

2. Network Diagnostics → It helps check wheather a destination is reachable

3. Testing Connectivity :- It helps check wheather a distination is reachable.

4. Route Information → Provides Information about routing problems.

ICMP Header format

Type (8 bits)
Code (8 bits)
Checksum (16 bits)
Data.

Fields

1. Type → Specifies the message type
2. Code → Gives additional information about the message
3. Checksum → Checks for errors.
4. Data → Contains message - specific information.

Advantages of ICMP.

- Detects network errors
- Helps troubleshoot network problems
- Used by Ping and Traceroute
- Improves network management